Quantitative Macro-Credit Risk Modeling

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Abstract

Most traditional credit risk models, primarily based on logistic regression, take the assumption that the relationship between the borrower's characteristics and the probability of default (PD) remains static over time. However, past systemic shocks such as the 2008 financial crisis and COVID-19 pandemic have shown that such an assumption does not hold because these models often fail to capture macroeconomic shifts, resulting in predictivity instability. This project proposes a state of the art regime-dependent credit risk model which captures conditioned probability of default (PD) on three frontiers: individual borrower features, real-time macroeconomic indicators and latent market regimes. Using high frequency dataset merged from Lending Club loan records and FRED macroeconomic series, the project employs Hidden Markov Model (HMM) and Gaussian Mixture Models (GMM) for economic state detection, Extreme Gradient Boosting (XGBoost) for predicting PD given borrower features, macroeconomic states and latent market regimes. Results from simulated data indicate that the macro-augmented and the regime-dependent models perform better with ROC-AUC of 0.8935 and 0.8947 compared to 0.5263 for the base model (uses borrower only features). This result suggests that incorporating macroeconomic conditions and market regimes improves the prediction of default rates.

Keywords: Credit Risk, Machine Learning, XGBoost, Hidden Markov Models, Macroeconomic Conditioning, Regime-Switching, Probability of Default.

1. Introduction

Credit risk models have long relied on logistic regression and scorecard-based approaches. These methods work reasonably well under normal conditions, but they share a fundamental assumption that tends to break down in practice: that the link between a borrower's characteristics and their likelihood of defaulting stays roughly constant over time.

The 2008 financial crisis and the COVID-19 shock made this assumption hard to defend. Default rates spiked in ways that static models simply didn't anticipate, not because the borrowers looked different on paper, but because the macro environment had shifted entirely. A model trained on 2005-2007 data had no way of knowing what a 10% unemployment rate or near-zero interest rates would do to repayment behavior.

What we're trying to do in this project is account for that. Instead of treating credit risk as a fixed function of borrower features, we want to condition it on the macroeconomic environment and on which "regime" the economy is in at a given point in time, whether that's a period of expansion, stress, or something in between.

The Fama-French model gave us a useful way to think about this. Just like equity returns can't be explained by market beta alone and benefit from adding size and value factors, credit risk probably can't be captured by borrower characteristics alone. There's a systematic component driven by macro conditions and market regimes that static models ignore.

So the framework we're building models the probability of default (PD) as a function of three things: the borrower's own features, macroeconomic variables at the time of assessment, and a latent regime label estimated from the data. The core question we want to answer is whether adding those last two components actually improves prediction in a meaningful way, not just on paper, but across different economic periods, including stress scenarios.

2. Research Questions and Objectives

This project follows the following key research questions:

1. Does incorporating macro-economic conditions in credit risk modelling improve the predictive performance of the probability of default of macro-economic conditions + base (borrower only) model versus the traditional borrower-only model?

2. Does the incorporation of latent economic regimes in credit risk modelling improve the performance of the model further, especially in times of economic turbulence?

3. Which regime shift detection methods best captures meaningful economic state for modelling in credit risk?

4. Do regime-aware models contribute to improvements in credit risk model calibration and stability of probability of default estimates across different economic cycles?

5. Does the proposed framework, i.e., regime-dependent stress-testing framework, generalize well in different economic cycles (calm, turbulent, recession, etc.)

The project aims to address these research questions by executing the following objectives:

1. Integrating and synchronizing multiple data sources, including the integration of the Lending Club loan dataset with FRED macroeconomic data series (e.g., Unemployment Rate). We achieve this using a point-in-time join method, ensuring no data leakage.

2. Implementing and comparing the Hidden Markov Model, Gaussian Mixture Models, and K-means clustering, and detecting the most statistically robust economic regimes.

3. Iteratively developing a three-tier modelling hierarchy using the Gradient Boosting (XGBoost) algorithm:

   Model 1: Borrower’s characteristics only (Traditional Model)

   Model 2: Borrower’s characteristics + macroeconomic features

   Model 3: Borrower + Macro + Regime-label features

4. Comprehensively assessing the performance of the three models using AUC and calibration curves to determine if the regime-awareness of a model decreases probability drift during periods of economic shift.

5. Performing stress testing simulation of extreme economic shocks such as the 2008 economic crisis and Covid-19, and evaluating the stability of probability of default estimates, and generalizing the results across different market conditions.

3.0 Literature Review

The 2008 financial crisis pushed credit risk modeling to the top of the research agenda in ways it had not been before. Banks lost enormous amounts of money, and a big part of the reason was that their models were not built to handle the kind of conditions that emerged. Since then, research has moved steadily toward approaches that can handle non-linear relationships and adapt to shifting economic conditions, rather than assuming the world stays roughly the same year after year.

This review covers three areas that directly shape this project. First, the classical credit risk models that still dominate practice and where they consistently fall short. Second, the growing body of work on connecting default probability to the broader macroeconomic environment. Third, the regime-switching literature, which addresses the fact that borrower behavior and default patterns do not look the same across all economic periods.

3.1 Traditional Credit Risk Models and Their Limits

Most discussions of credit risk in the academic literature start with Merton (1974). His structural model frames default as the point where a company's asset value drops below what it owes a clean piece of theory, but one that depends on inputs that cannot be directly observed in practice. That gap between theory and application pushed researchers toward reduced-form models, notably Jarrow and Turnbull (1995) and Duffie and Singleton (1999), which treat default as a probabilistic event governed by a hazard rate rather than something derived from unobservable asset dynamics. These caught on in practice but came with their own limitations: they still assume the relationship between a borrower's characteristics and their default probability does not change much over time.

In consumer lending, logistic regression has been the default tool for decades. Altman's Z-score (1968) is the most cited version of this idea, a fixed weighted combination of financial ratios used to classify firms as distressed or not. The model has been updated and extended many times, but the underlying structure has stayed the same: a linear mapping from inputs to risk, estimated once and applied going forward. Bello (2023) compared these classical methods to more recent machine learning alternatives and found logistic regression holds up well when data is limited but gets overtaken when non-linear patterns are present. Suhadolnik et al. (2023) frame the problem more bluntly: the real issue is not that the model is too simple, it is that it treats economic conditions as irrelevant, which they argue is a structural flaw rather than a tuning problem.

3.2 Machine Learning Applied to Credit Risk

Ensemble methods started getting serious attention in credit risk around 2015. Lessmann et al. (2015) compared 41 classifiers across multiple datasets and found that boosting methods consistently deliver stronger AUC scores than classical alternatives. They were careful to flag something important though strong AUC does not guarantee well-calibrated probabilities, and in credit risk that distinction matters. A lender needs PD estimates that reflect actual default rates, not just a model that ranks borrowers correctly relative to each other. Brown and Mues (2012) made a complementary point about gradient boosting specifically: it handles missing values and feature interactions better than logistic regression and does not require the extensive manual feature engineering that traditional scorecards demand.

One pattern worth noting across this literature is how often the time dimension gets ignored in model evaluation. Random train-test splits are still common practice in published studies, which can produce results that look good on paper but would not hold up in deployment. A model trained on data from 2015 and evaluated on data from 2010 has effectively been given information it could not have had at the time. This project avoids that by using a strict time-based split throughout, with evaluation only on loans issued after a fixed cutoff date.

3.3 Macroeconomic Conditioning in Credit Models

Connecting credit risk to macroeconomic conditions is not a new idea, but embedding it properly into a machine learning pipeline is still relatively underdeveloped. Wilson's CreditPortfolioView (1997) was one of the earlier frameworks to do this systematically, modeling portfolio-level default rates as a function of GDP growth, unemployment, and interest rates through a system of equations. It showed that macro variables carry real explanatory power for default behavior at the aggregate level, even if firm-level characteristics dominate at the individual borrower level.

The regulatory framework reflects this too. Basel II and III draw a formal distinction between through-the-cycle and point-in-time PD estimates (BCBS, 2006). Through-the-cycle estimates try to average out cyclical variation; point-in-time estimates reflect where risk stands given current economic conditions. A model that conditions on macro variables is essentially doing the latter; it is trying to capture how risk shifts as the economy moves through different phases, rather than treating the cycle as noise to be smoothed away.

The choice of macro variables in this project is grounded in empirical findings. Figlewski et al. (2012) found that unemployment and industrial production growth have statistically significant effects on corporate default rates, holding firm-level factors constant. Blanco et al. (2020) showed that high-yield credit spreads including the BAMLH0A0HYM2 series used here carry forward-looking information about aggregate default risk that borrower data alone does not reflect. Both findings point toward the same conclusion: leaving macro variables out of a credit model means leaving out information that genuinely moves default rates.

3.4 Regime Switching Framework

Hamilton (1989) introduced Hidden Markov Models to economics through a deceptively simple observation: US GDP growth does not behave as if it were drawn from a single stable distribution. It behaves more like a process that switches between two states expansion and contraction with the active state hidden from direct observation but inferable from the data. That idea opened up a whole line of research into latent regime models for financial and economic time series.

The implications for credit risk are significant. Nickell et al. (2000) showed that credit rating transition probabilities vary substantially across economic states. A borrower sitting at a given rating is meaningfully more likely to be downgraded during a recession than during an expansion, even if their underlying financials look similar. A model that assumes these probabilities are constant is going to get things systematically wrong every time the economic environment shifts.

Bai et al. (2019) brought this logic into a machine learning context by training separate gradient boosting models for each regime identified by a two-state HMM. Their regime-specific models outperformed a single pooled model on both AUC and Brier score, which is probably the most direct empirical precedent for what this project does. The approach here extends theirs in a few ways: three regimes instead of two, multiple clustering methods compared rather than HMM alone, and the regime label included as a feature in the model rather than used to partition the training data entirely which keeps the model from losing information during regime transitions.

Gaussian Mixture Models are also included as an alternative to HMMs. Frühwirth-Schnatter (2006) makes the case that GMMs can be more appropriate when there is no strong reason to assume regime transitions are Markovian. Whether that assumption holds for the specific macro feature set used here is an open empirical question, and comparing the two methods directly is part of how this project tries to answer it.

4.0 Methodology & Approach

In this project, we propose a regime-dependent credit risk model that incorporates borrower features, macroeconomic conditions at a given time t, and latent market regimes to evaluate the creditworthiness of a borrower by assessing the probability of default of the borrower. We implemented a multi-step quantitative approach, moving from synthesis of raw data to unsupervised regime detection (via Hidden Markov Model), and finally to supervised predictive modeling (via Extreme Gradient Boosting). The guiding equation is:

$$PD = Pr(default \mid X_{borrower}, M_t, R_t)$$

$X_{borrower}$: borrower-specific characteristics

M_t: macroeconomic variables at time t

R_t: latent market regime at time t

4.1 Data Synthesis

Our data layer comprises two primary data sources:

Micro-level data (Loan Lending data), which we obtained from Lending Club, represent comprehensive peer-to-peer loan records. This dataset was the primary record for borrower features, including FICO score, Debt-to-Income Ratio, employment length, and loan status, i.e., default vs. non-default. This data set formed the basis for understanding the behavior of a borrower, which is key when analysing the credit risk and making decisions on whether to approve or decline a loan request from borrowers(cite). **Macro-level data**, high-frequency data, were obtained from FRED. This data captures the economic conditions, such as market stress (High-Yield Spread, proxy for market stress), systemic risk, consumer health (proxied by unemployment rates, consumer price index, etc), industrial production index, interest rates (proxied by federal funds effective rates), and real gross domestic product. We implement a point-in-time join to merge micro and macro-level data i.e., macro-indicators with loan records on the basis of loan issuance time. This ensures that the model only makes use of economic data available at the time of credit evaluation to avoid data leakage. The equation for achieving this is shown below:

$D_{merged} = \{(x_{i,t}, m_\tau) | \tau = \max(t_{macro} \leq t_{loan})\}$

Where:

$x_{i,t}$ – features for the borrower i at the loan issuance time t

m_τ – microeconomic indicator at time τ

t_{loan} – time stamp of credit decision

The above equation is a data filtering function that ensures that there is no look-ahead bias.

4.2 Regime Detection via Hidden Markov Model, HMM

Before training our regime-aware predictive model, we implemented Hidden Markov Model with Gaussian emission to help identify latent economic states. The Hidden Markov Model was trained on macroeconomic indicators with a monthly frequency, and the expected output was discrete regime labels, **R_t = {0, 1, 2},** which represent expansion, transition, and stress states of the economy for every period in the data set.

The equation below gives the probability of observing a sequence of macroeconomic indicators, O, given latent economic regimes, Q:

$P(O|\lambda) = \sum_Q \pi_{q1} b_{q1}(O_1) \prod_{t=2}^{T} a_{qt-1} b_{qt}(O_t)$

Where:

$a_{ij} = P(q_t = S_j | q_{t-1} = S_i)$ – transition probability between states

$b_j(O_t) \sim N(\mu_j, \Sigma_j)$ - Gaussian emission probability for the state j

$S_t \epsilon \{0, 1, 2\}$ – discrete regime labels assigned to period t

4.3 Supervised Modeling Framework

We utilized the Extreme Gradient Boosting model primarily due to its effectiveness in handling non-linear interactions and missing data more efficiently. We iteratively modelled credit risk starting with a baseline model, trained on specific borrower characteristics, X. The XGBoost loss function is defined as:

$\mathcal{L}(\phi) = \sum_i l(\hat{y_i}, y_i) + \sum_k \Omega(f_k)$

Thus, the baseline model is defined by the equation:

$$PD = \hat{y} = f(X_{borrower}) = Pr(default \mid X_{borrower})$$

We then geared up to the Macro-augmented model, which combines both the borrower characteristics, X, and macro-economic conditions, m_t. The equation for this model is defined as:

$PD = \hat{y} = f(X_{borrower}, m_t) = Pr(default \mid X_{borrower}, m_t)$

Finally, we incorporated latent regime in our state-of-the-art regime-dependent credit risk model, allowing the model to learn split point for different regimes, borrower features, and economic state. This model is defined by the equation:

$$PD = \hat{y} = f\left(X_{borrower}, m_t, R_t\right) = Pr\left(default \mid X_{borrower}, m_t, R_t\right)$$

In each of these models, the output is a probability measure of the likelihood of a borrower defaulting on their loan obligation, conditioned on their past behavior, past behavior and macroeconomic conditions, and past behavior + macroeconomic conditions + market regimes.

4.4 Model Evaluation and Stress Testing

We performed model evaluation using the precision-recall AU and ROC-AUC metrics as the primary model evaluation frameworks. We also used the Brier Score and calibration curves to evaluate the model calibration of probability of default estimates. We also performed out-of-sample stress testing by conducting a crisis simulation, which was achieved by shifting the macro-input features manually to the 2008 financial crisis level and then observing the sensitivity and migration of probability of default scores across the models

Figure 1 below provides a visual of the workflow of our project from developing data pipeline to stress testing and scenario analysis.

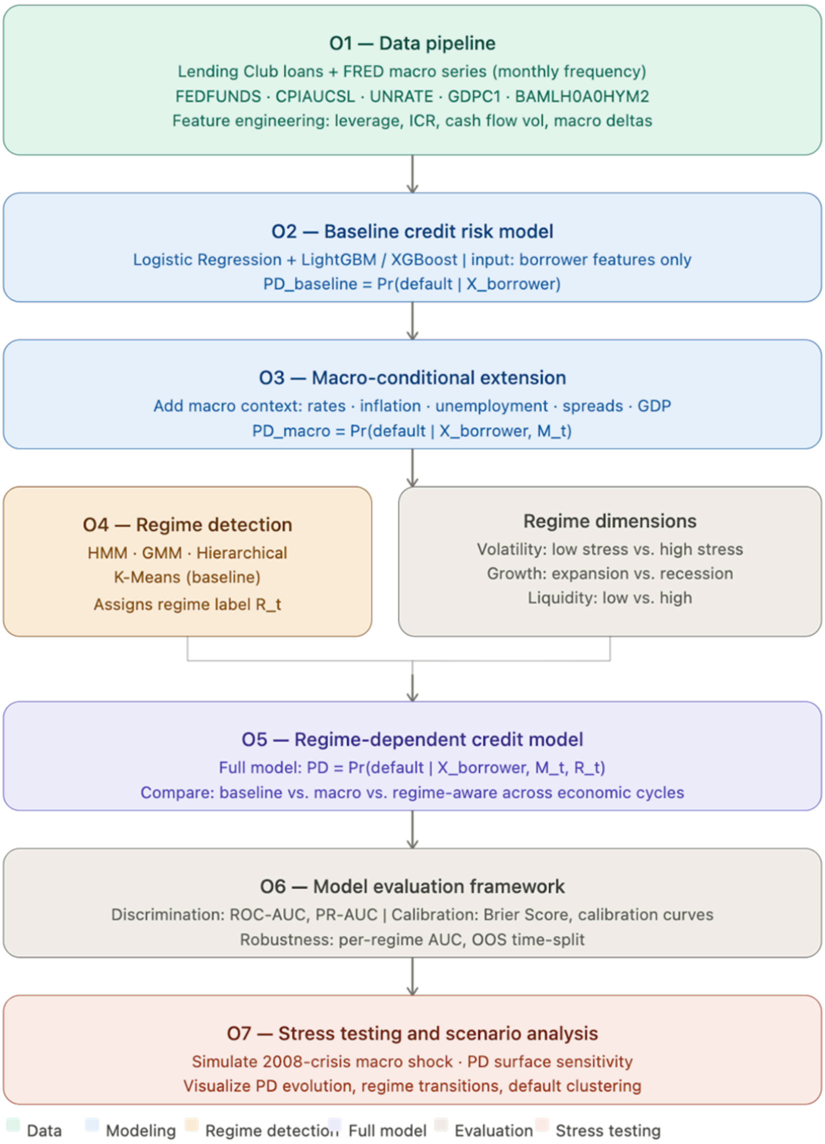


Figure 1

5.0 Results

6.0 Discussion & Conclusion

7.0 Future Direction

Appendix

GitHub and Code Repository

All project code is hosted publicly at:

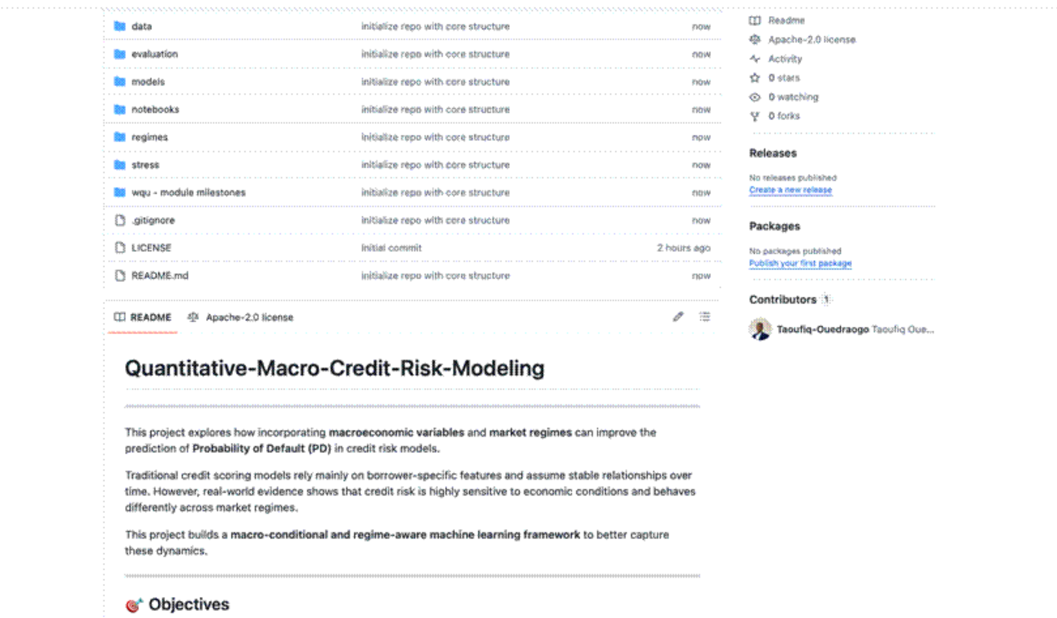
https://github.com/Taoufiq-Ouedraogo/Quantitative-Macro-Credit-Risk-Modeling

The repository has been initialized and structured in advance of the implementation work, which begins in Module 4. At this stage, the folder architecture is in place and reflects the project objectives directly.

We organized as follows:

- *data/* : scripts for gathering Lending Club loan dataset with FRED macroeconomic series
- *models/* : implementations of our 3 model: borrower-only baseline + macro-conditional extension + full regime-dependent model
- *regimes/* : unsupervised regime detection using HMM, GMM, K-Means and hierarchical clustering, along with logic for assigning regime label to each observation
- *evaluation/* : structured evaluation scripts covering ROC-AUC, PR-AUC, Brier score, calibration curve and the time-based out-of-sample split
- *stress/* : 2008-crisis macro shock simulation + PD surface visualization across macro states and regime transitions
- *notebooks/* : exploratory data analysis and interim results as they become available

Code development, initial results will be reported starting from Module 4. The repo will be updated continuously from that point forward, and access will be shared with the instructor as required.



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Acknowledgement

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